

The Minimum Viable Brain

Evolving adaptive behavior through training perturbations in minimal embedded neural networks

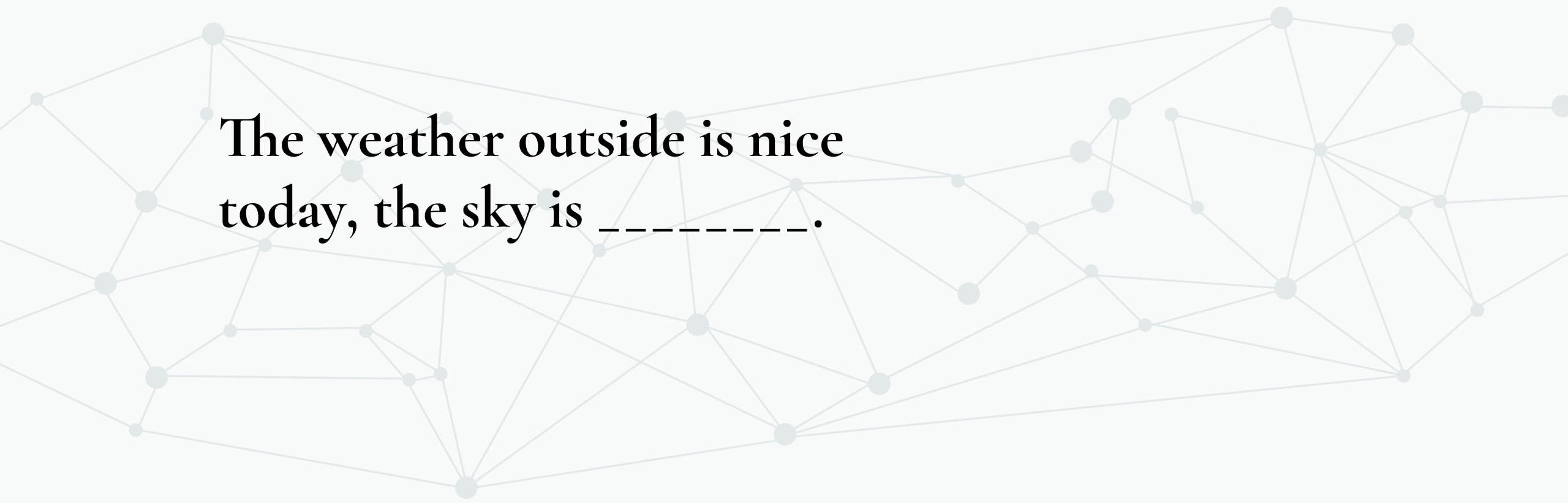
Dr. Jana Katharina Dahlmanns



1 - Who here knows how to work a coffee maker?

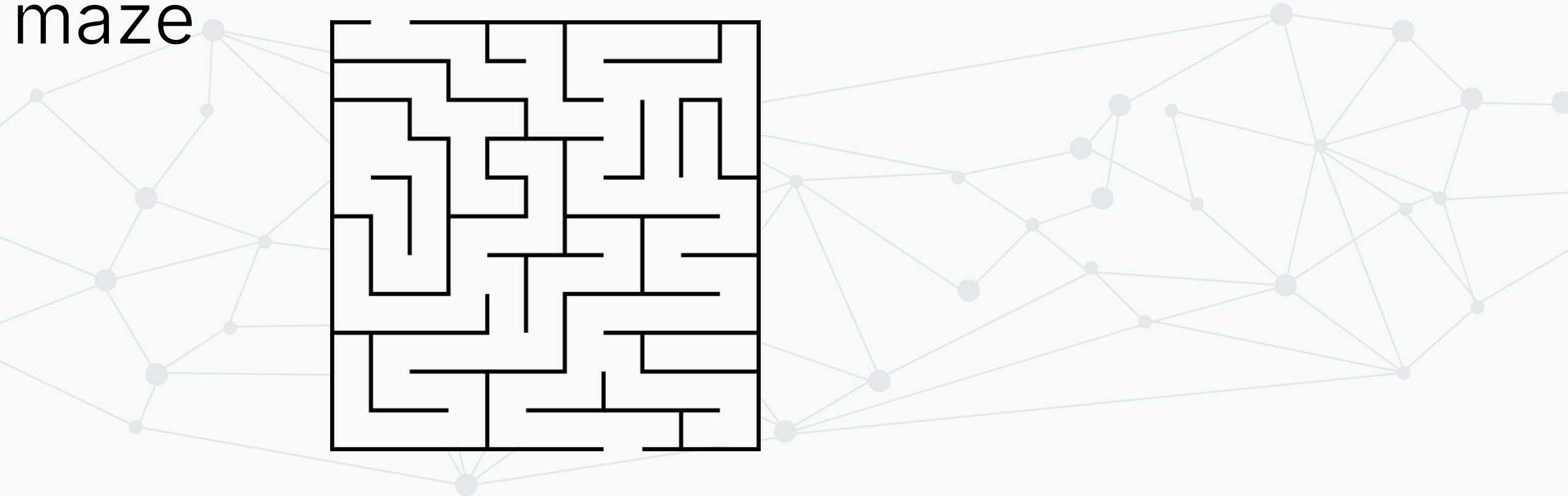


2 – Complete this sentence:



The weather outside is nice
today, the sky is -----.

3 – Find the shortest path through this maze



Why are brains so good at **flexible generalization?**

Generalization

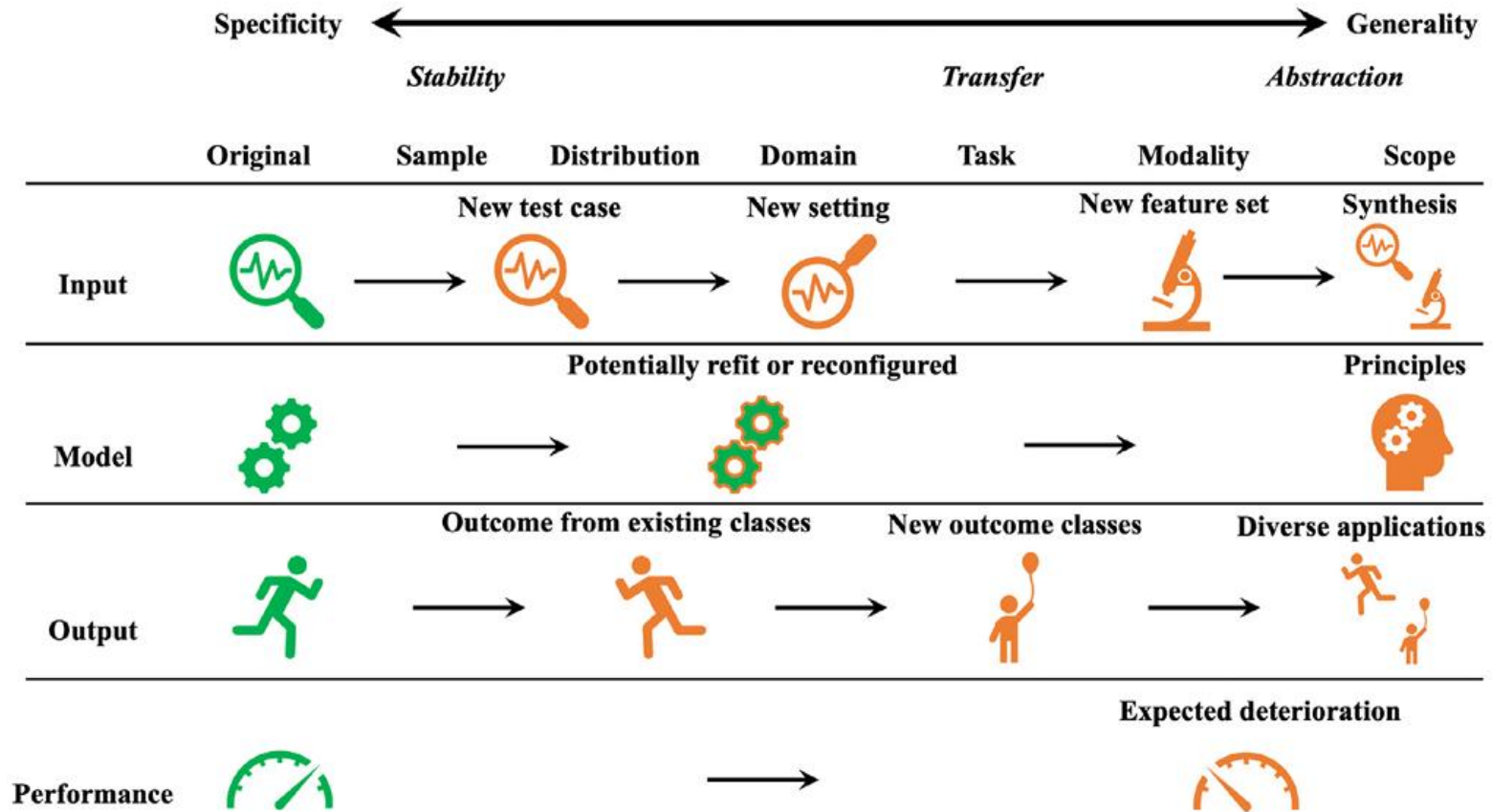


Fig. 1. Types of generalization.

Rohlf (2025) *Generalization in neural networks: A broad survey.*
Neurocomp

Why Computational Neuroethology?

“

Studies of mechanisms of adaptive behavior generally focus on neurons and circuits.

*But adaptive behavior **also depends on interactions among the nervous system, body and environment:***

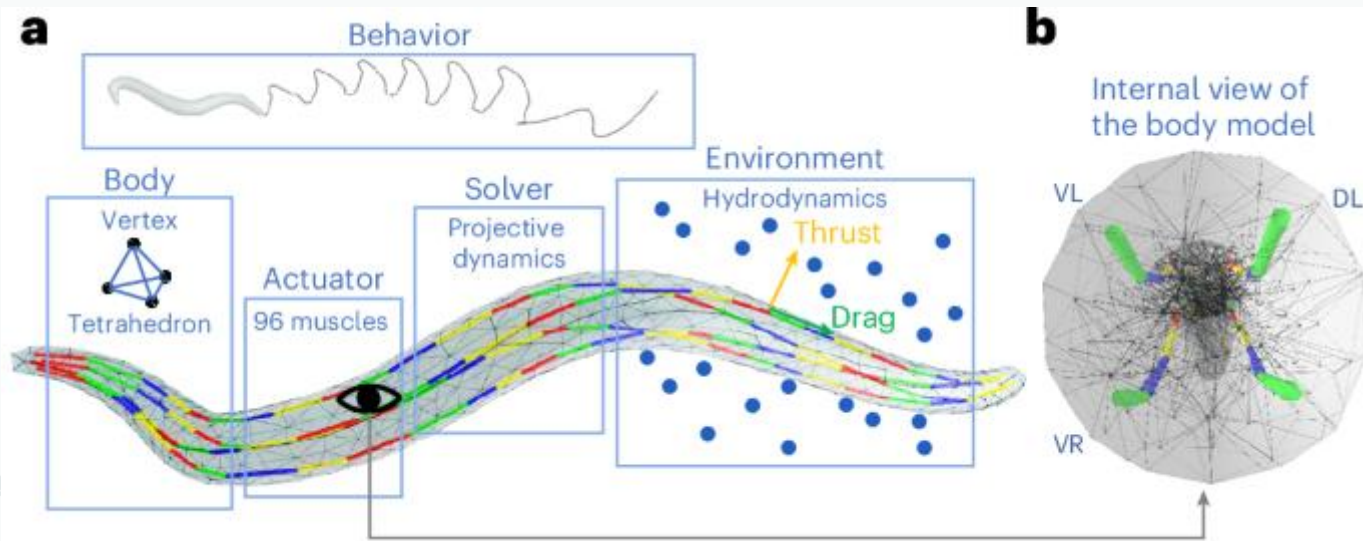
*sensory preprocessing and motor post-processing filter inputs to and outputs from the nervous system [...] and **continuous feedback between nervous system, body and environment are essential for normal behavior.** [...]*

Computational neuroethology, which jointly models neural control and periphery of animals, is a promising methodology for understanding adaptive behavior.

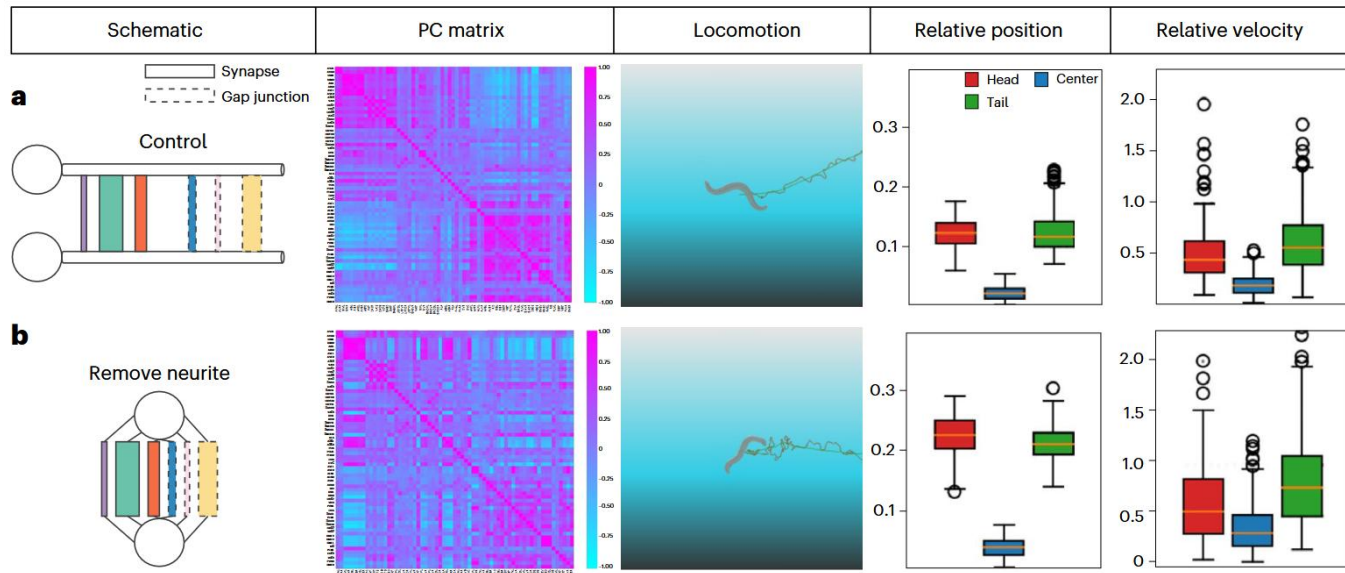
”

Hillel Chiel & Randall Beer (1997) *The brain has a body: adaptive behavior emerges from interactions of nervous system, body and environment.* Trends Neurosci

Models of Minimal Cognition

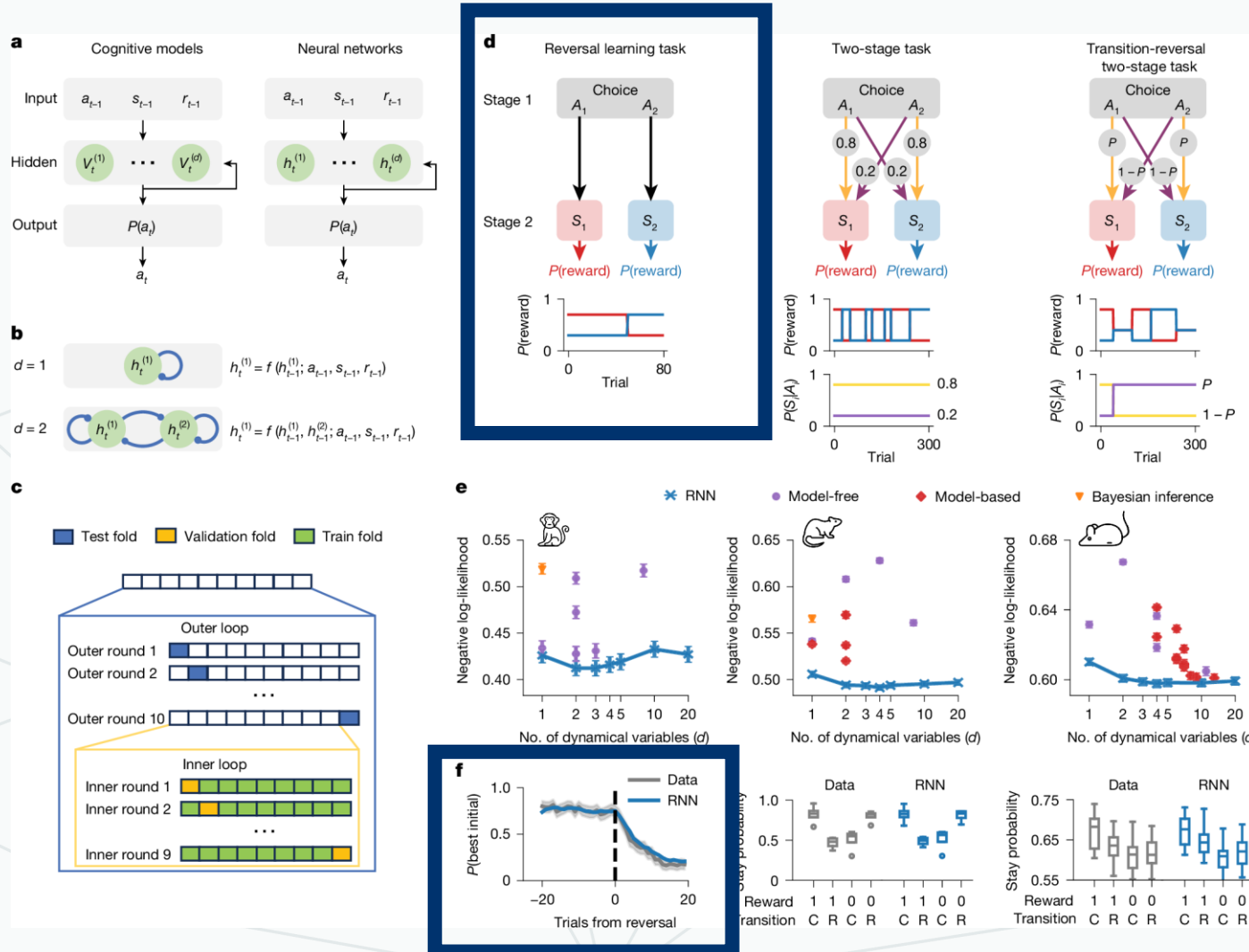


Embedding models into interactive environments can reveal neuronal dynamics and biologically realistic behavior not captured in isolated networks.



Zhao, M., Wang, N., Jiang, X. et al. (2024) An integrative data-driven model simulating *C. elegans* brain, body and environment interactions. Nat Comput Sci

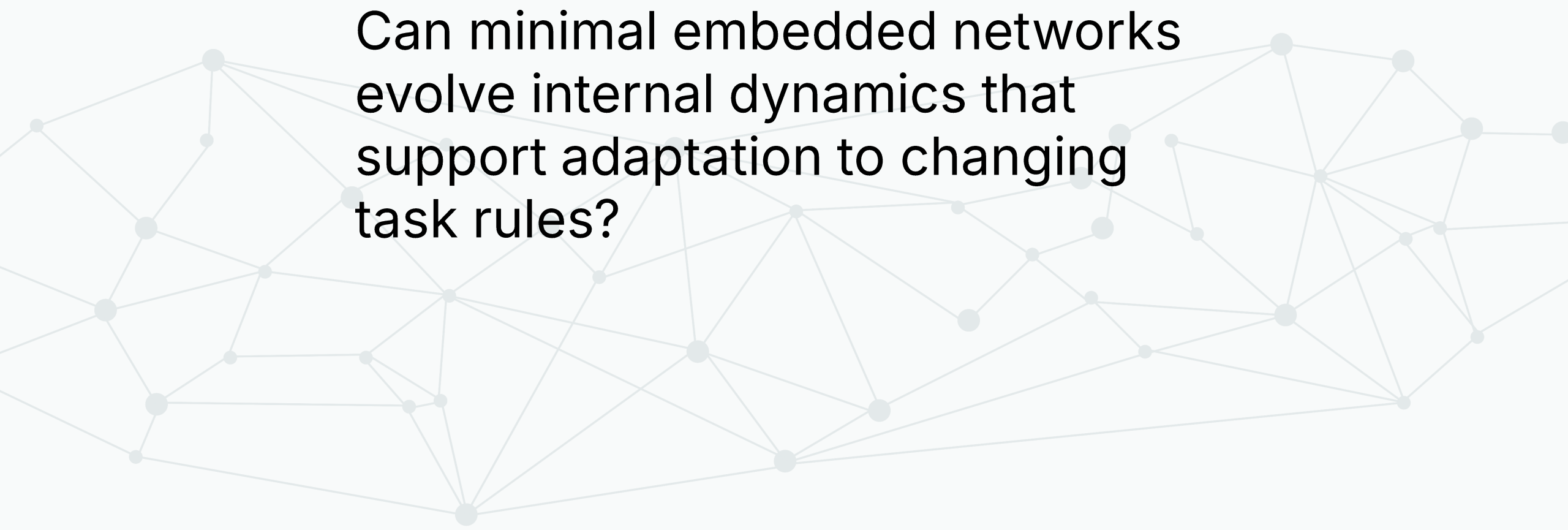
Models of Minimal Cognition



Tiny RNNs can capture complex behavior while remaining interpretable.

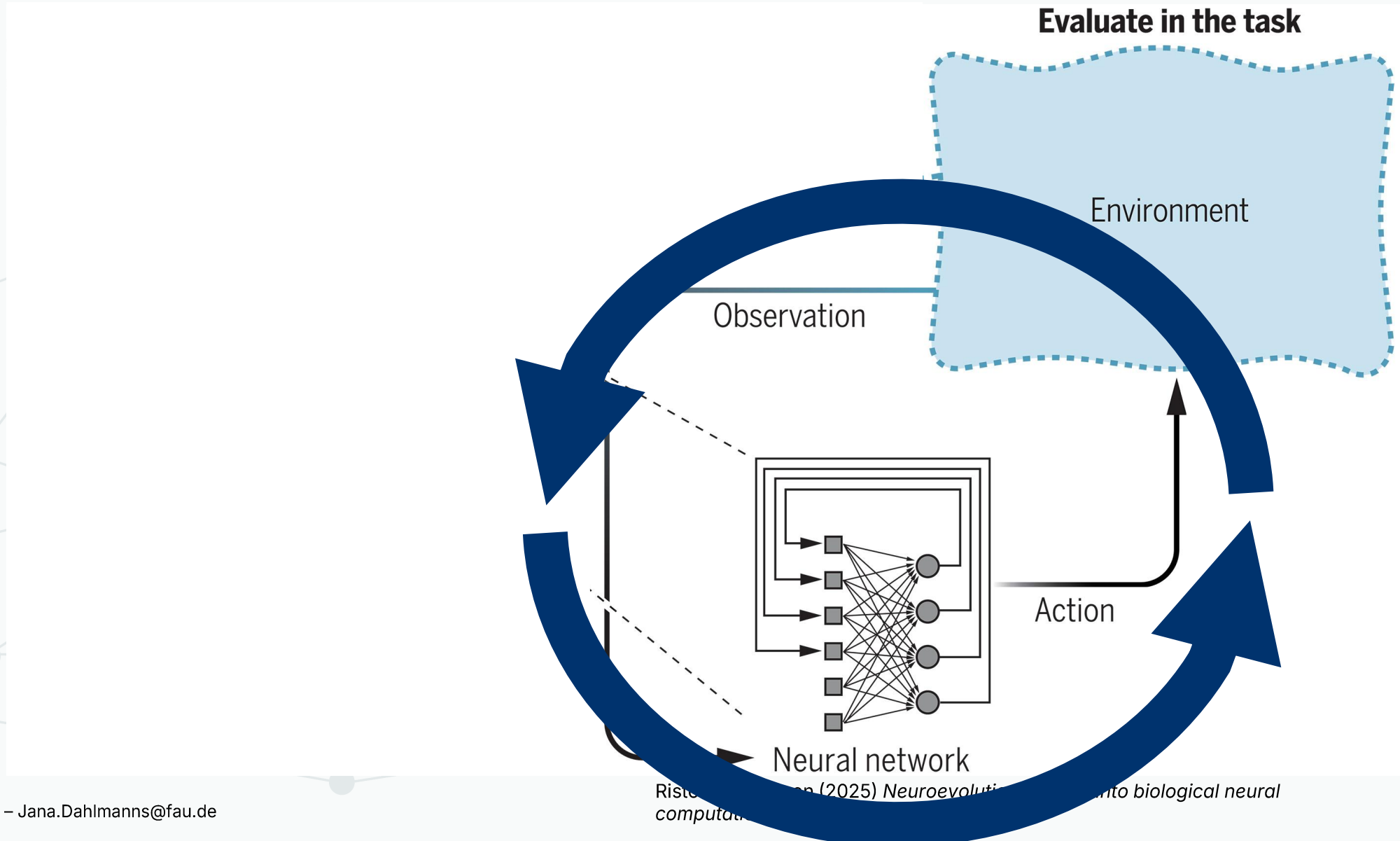
Ji-An, L., Benna, M.K. & Mattar, M.G. (2025) *Discovering cognitive strategies with tiny recurrent neural networks*. Nature

OUR QUESTION



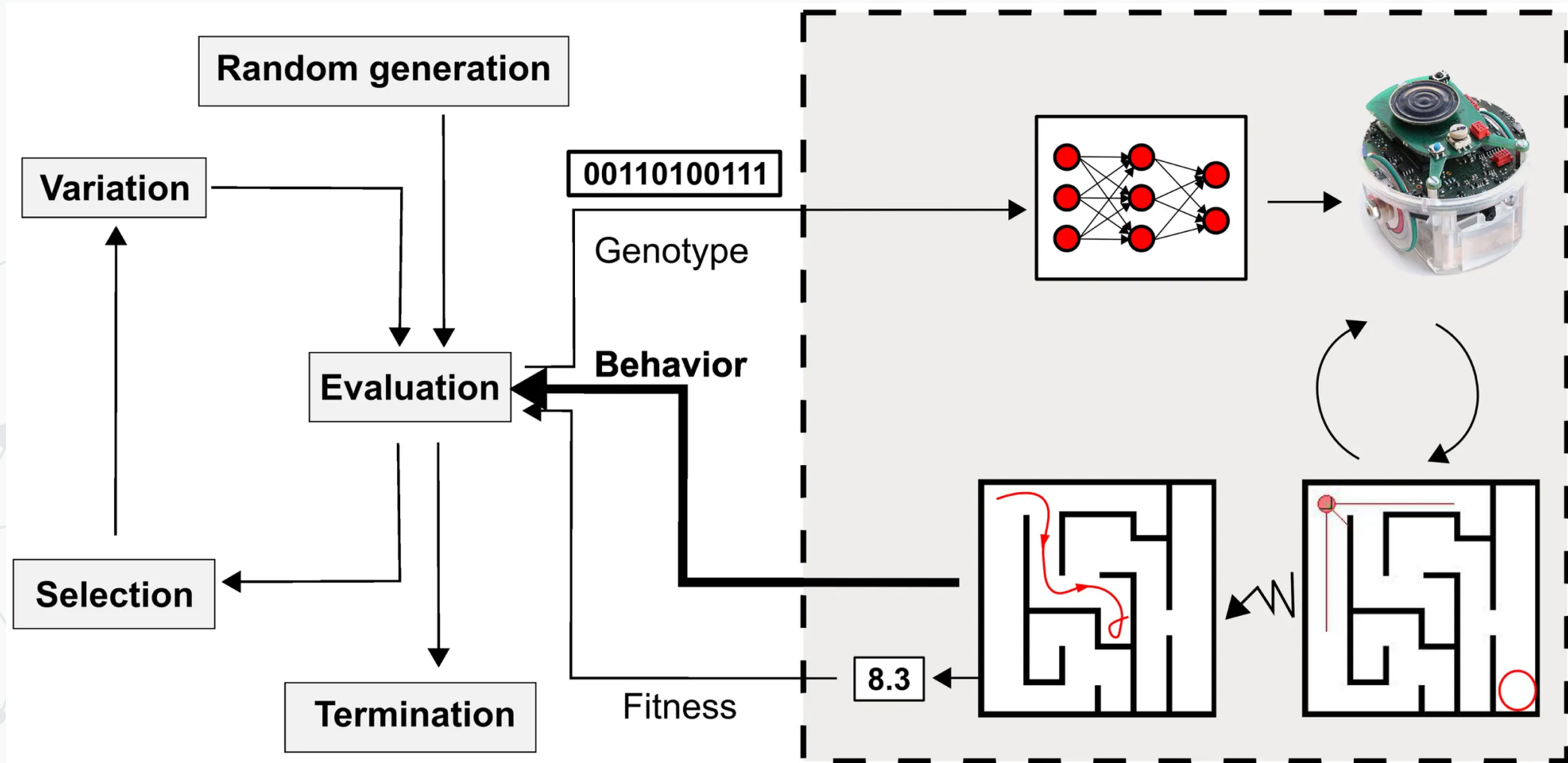
Can minimal embedded networks
evolve internal dynamics that
support adaptation to changing
task rules?

Evolutionary Agents



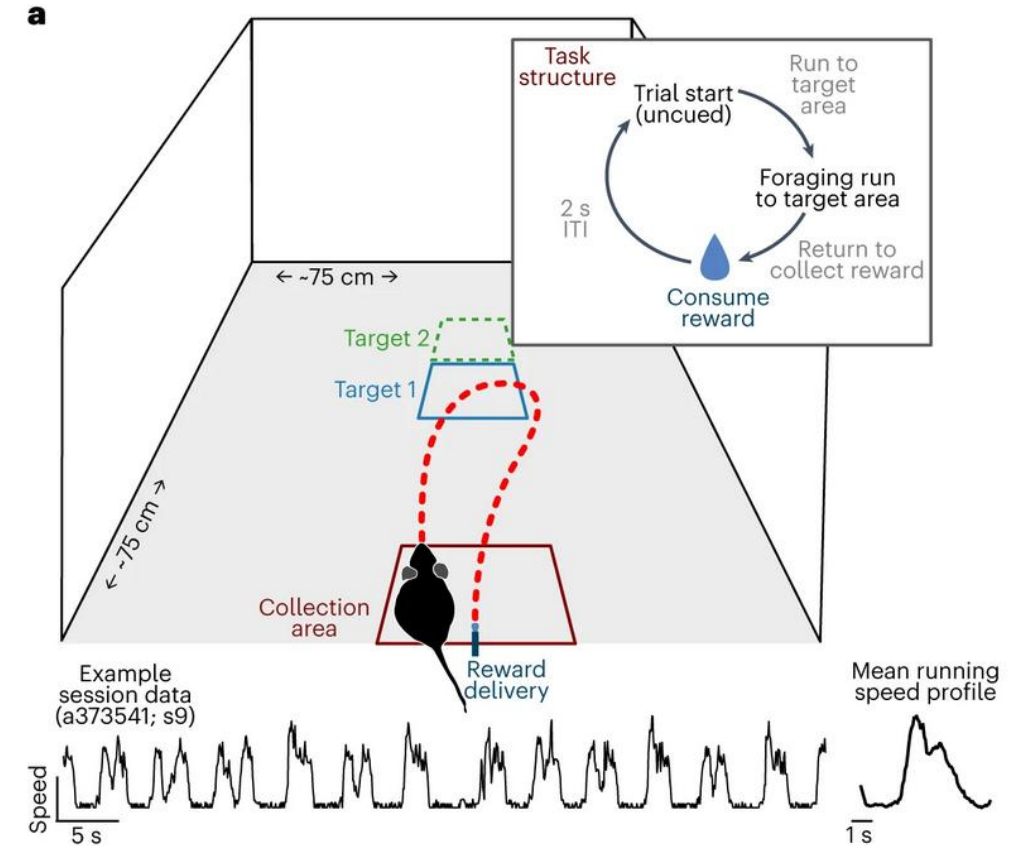
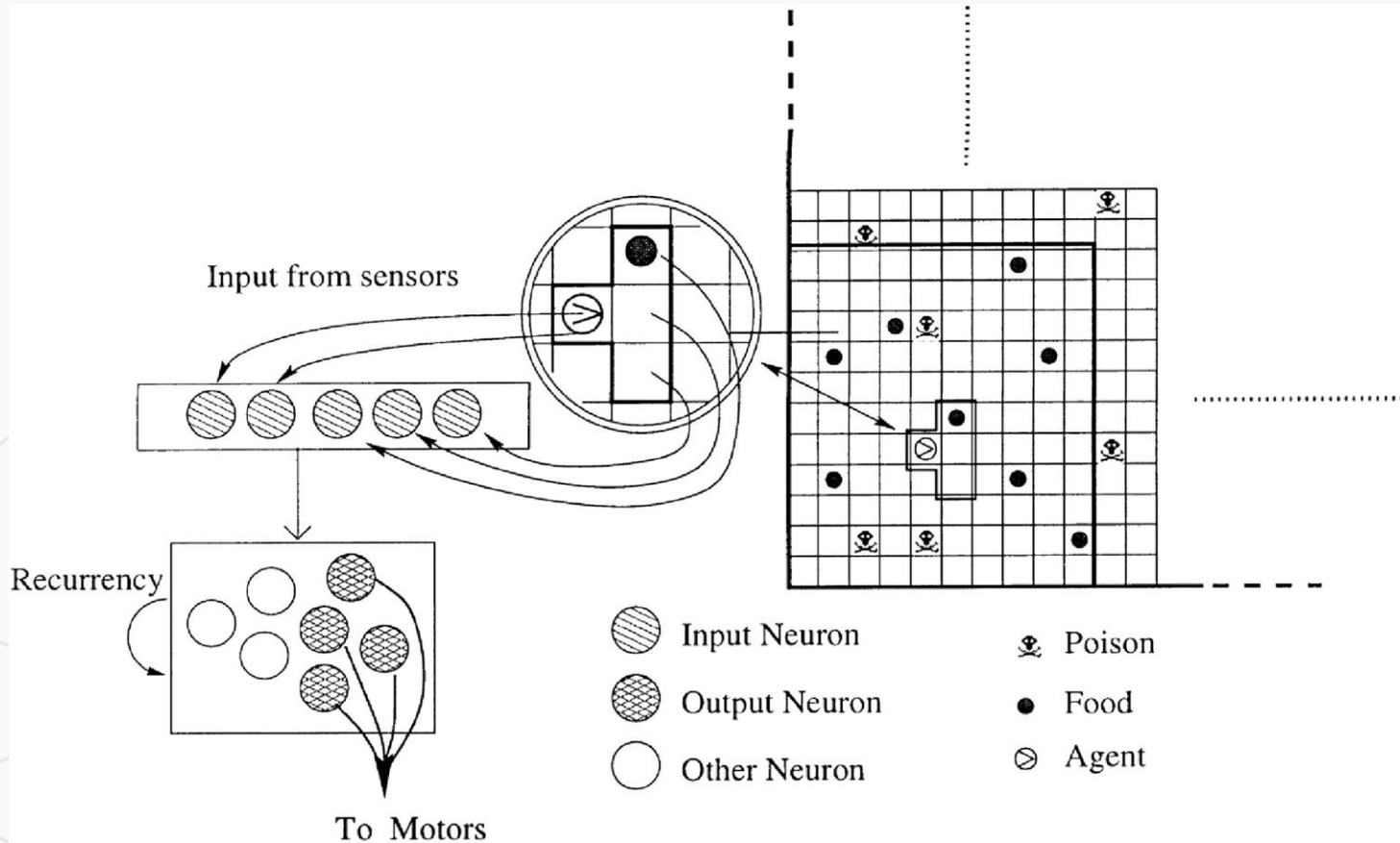
Risto et al. (2025) Neuroevolution into biological neural
computations

Evolutionary Algorithms



Doncieux S, Bredeche N, Mouret J-B and Eiben AEG (2015) *Evolutionary Robotics: What, Why, and Where to*. Front. Robot. AI

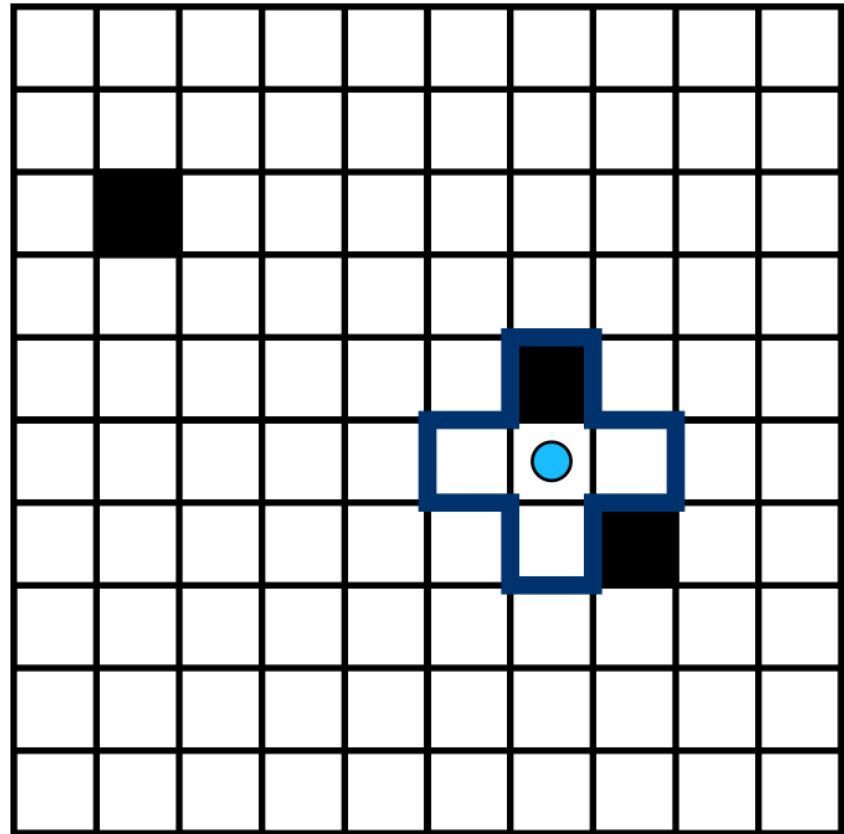
Foraging Task



Jiang, WC., Xu, S. & Dudman, J.T. (2022) Hippocampal representations of foraging trajectories depend upon spatial context. Nat Neurosci

Risto Miikkulainen (2025) Neuroevolution insights into biological neural computation. Science

Foraging Task



● Agent ■ Target + Sensory Input

No poison

Sensory input: 4 directions + current field

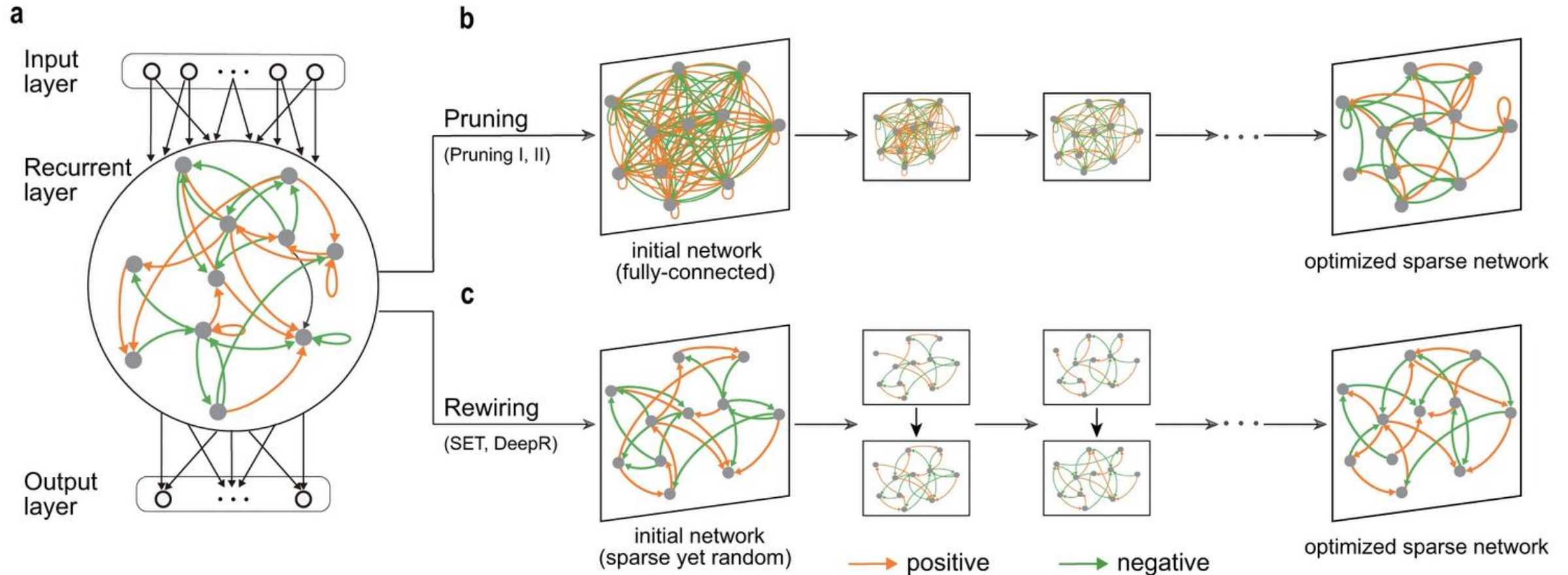
Output: movement in 4 directions + stay signal to forage

Task:

Metabolic cost + Movement cost

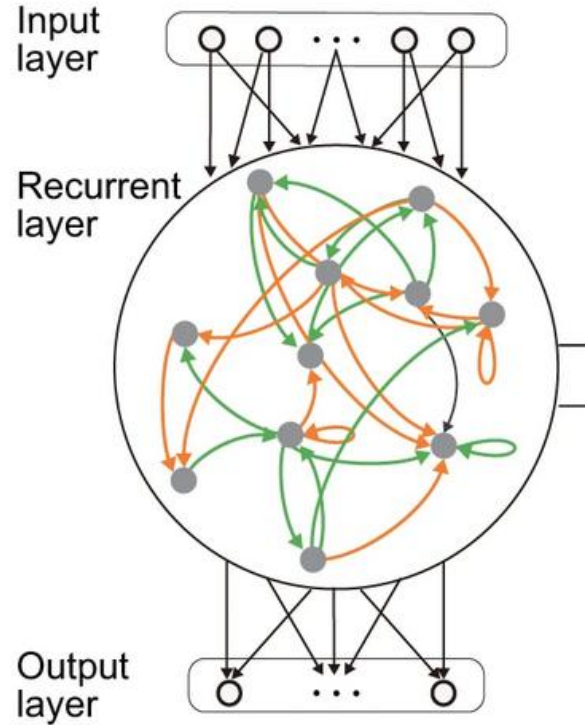
Foraging = energy refill

Neuronal Network – The agent's brain

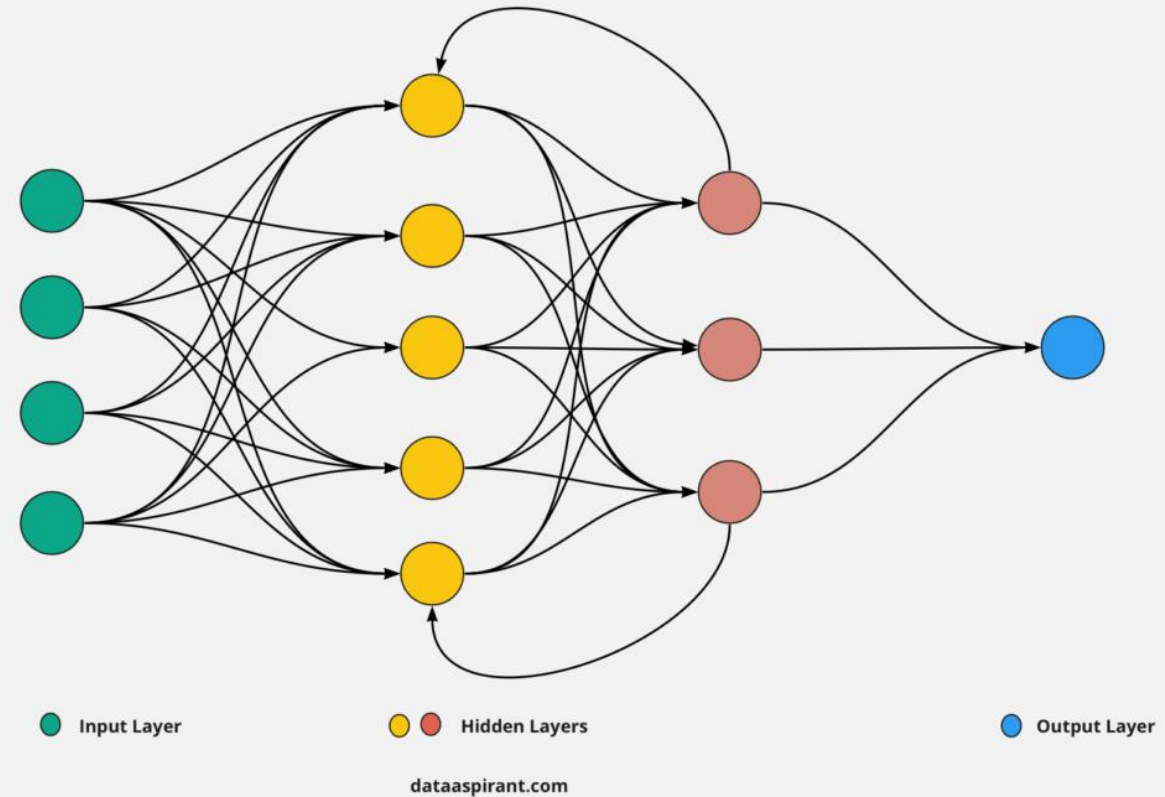


Zhang, X.J., Moore, J.M., Yan, G. et al. (2023)
*Universal structural patterns in sparse
recurrent neural networks.* Commun Phys

Neuronal Network – The agent's brain

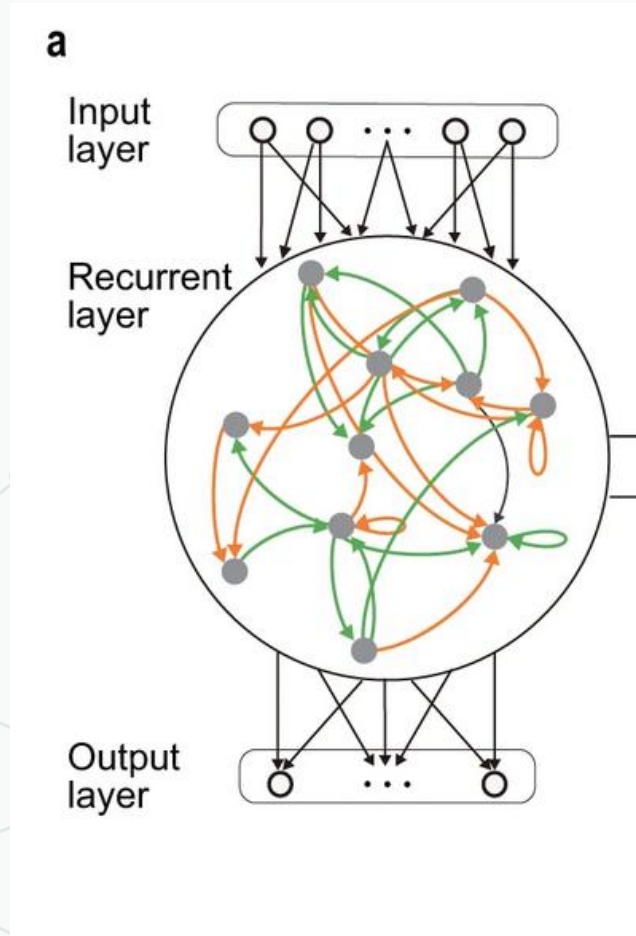
a

Recurrent Neural Network

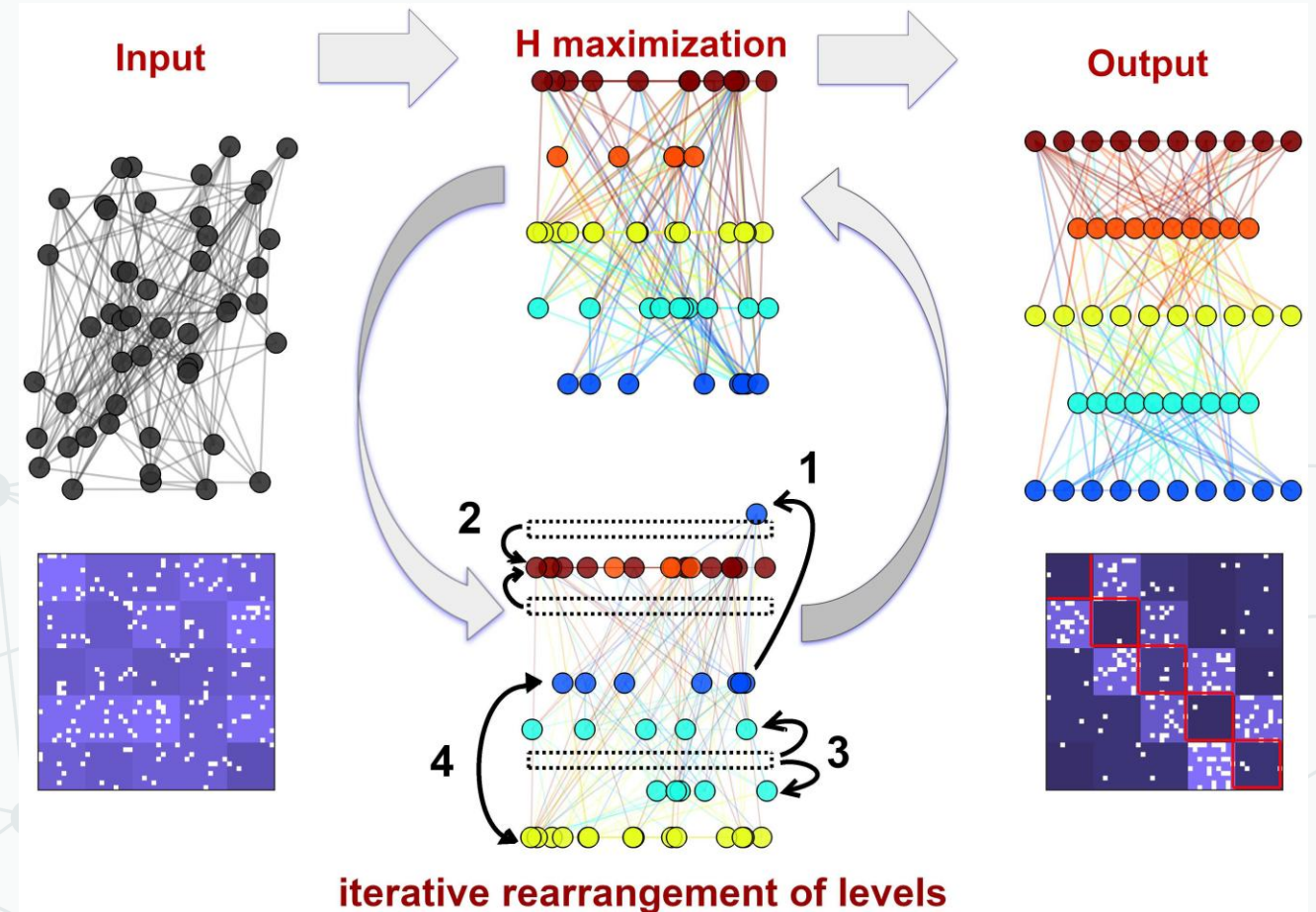


Zhang, X.J., Moore, J.M., Yan, G. et al. (2023)
*Universal structural patterns in sparse
recurrent neural networks.* Commun Phys

Unstructured Neuronal Networks



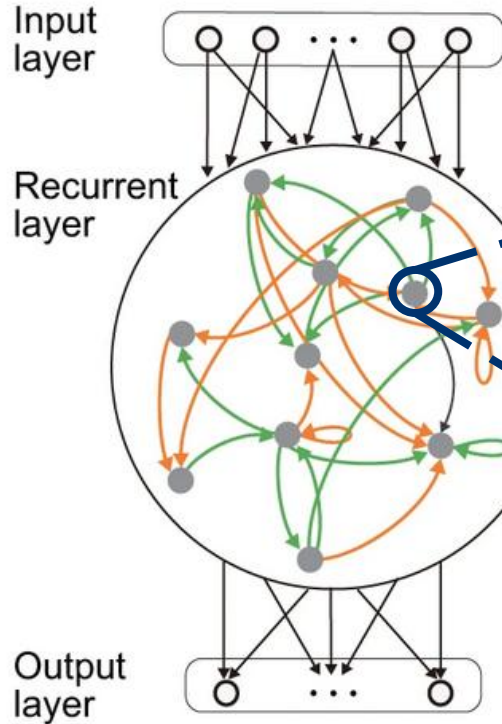
Zhang, X.J., Moore, J.M., Yan, G. et al. (2023)
Universal structural patterns in sparse recurrent neural networks. Commun Phys



A. Pathak, S.N. Menon, & S. Sinha (2024) *A hierarchy index for networks in the brain reveals a complex entangled organizational structure,* Proc. Natl. Acad. Sci. U.S.A.

The neurons

a



$$u_i(t) = \underbrace{b_i}_{\text{Tonic activation bias}} + \underbrace{\sum_j w_{ij} a_j(t)}_{\text{Weighted sum of inputs}} + \underbrace{\xi_i(t)}_{\text{Random normal noise}}$$

$$a_i(t+1) = \Theta(\tanh(u_i(t)) - \theta_i)$$

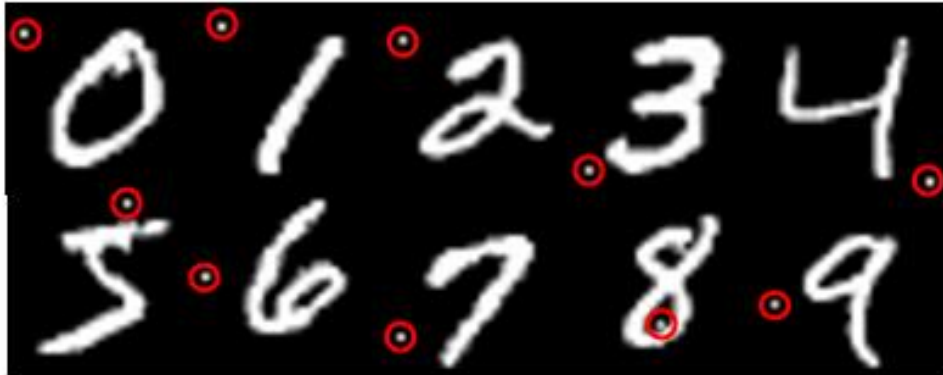
Threshold

**Meaningful
threshold mapping**

Zhang, X.J., Moore, J.M., Yan, G. et al. (2023)
*Universal structural patterns in sparse
recurrent neural networks.* Commun Phys

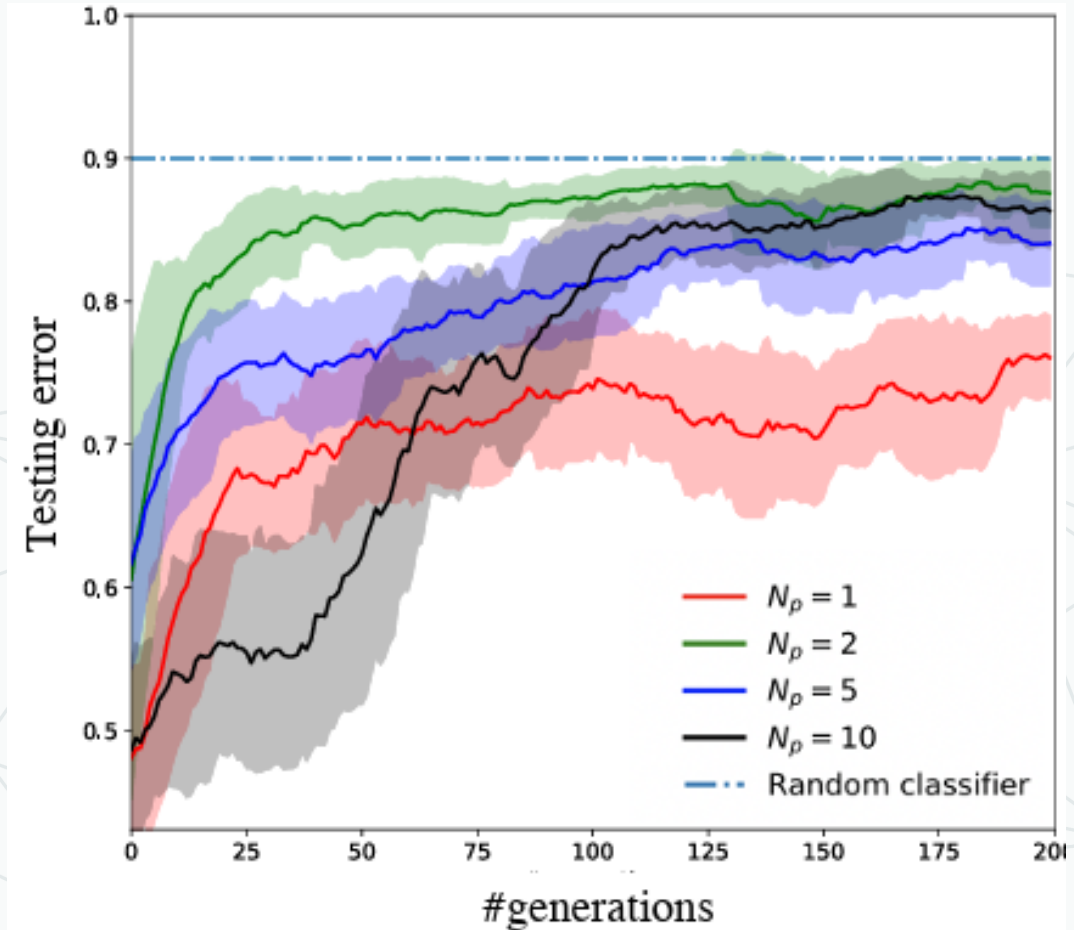
Candadai M. et al. (2020) *Universal structural
patterns in sparse recurrent neural networks.*
Commun Phys

Generalization through training perturbations



(a) Corrupted MNIST images

Chaudhury S., Yamasaki, T. (2020)
*Investigating generalization in neural networks
under optimally evolved training perturbations.*
arXiv



(c) Learning curve: MNIST

1

Static Task Phase

Train on static task
w/ traditional
evolution and define
adaptability gap

2

Training & Testing EA with perturbations

Train variants w/
perturbed evolution and
test performance and
adaptability

3

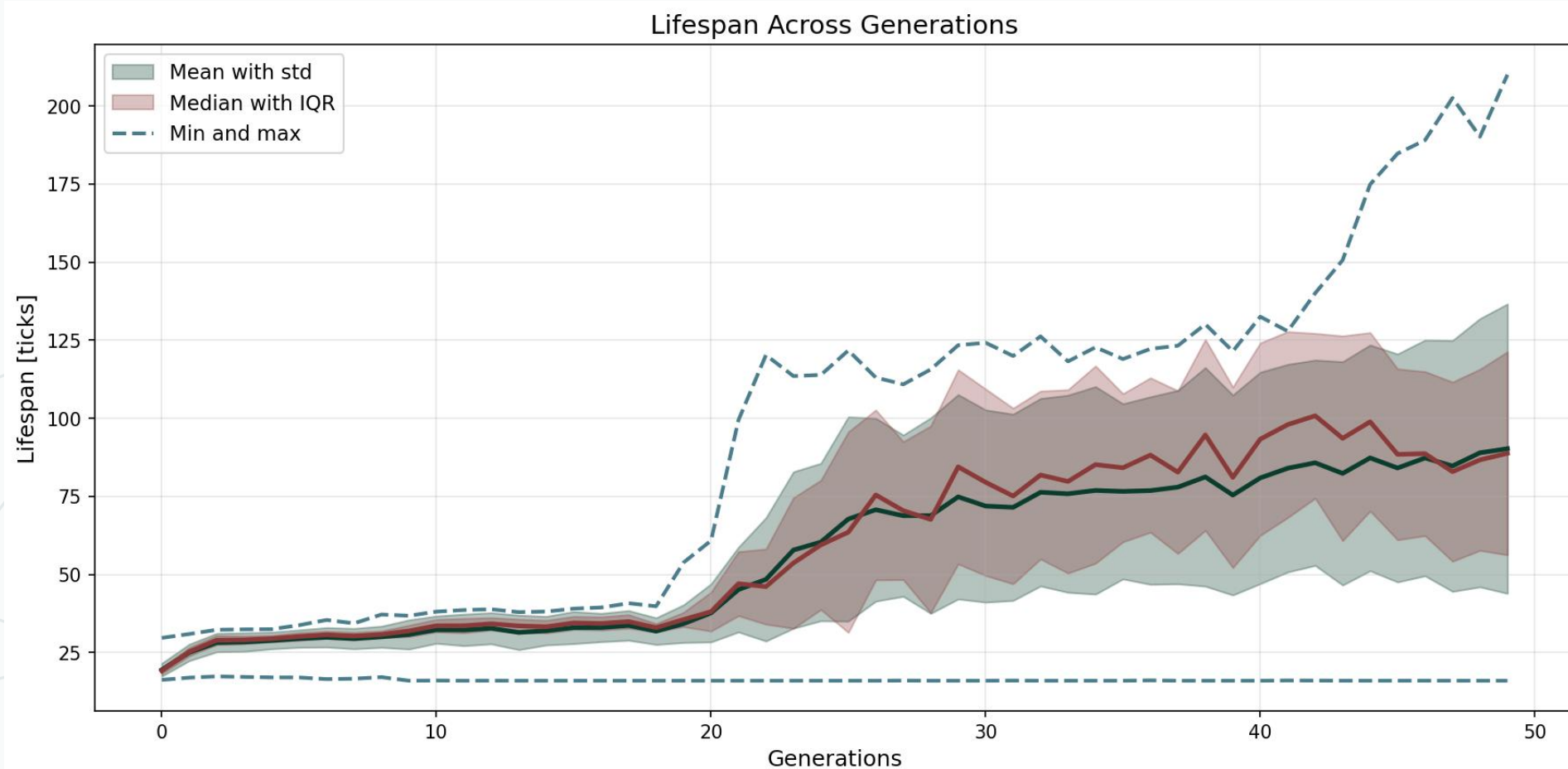
Ablation

Ablate to necessary
core and test
adaptability on withheld
scenarios

4

Wiring analysis

Analyze common
patterns in adaptors and
compare with biological
wiring patterns



EA evolved from random network initialization. Population size = 250, elite = 10

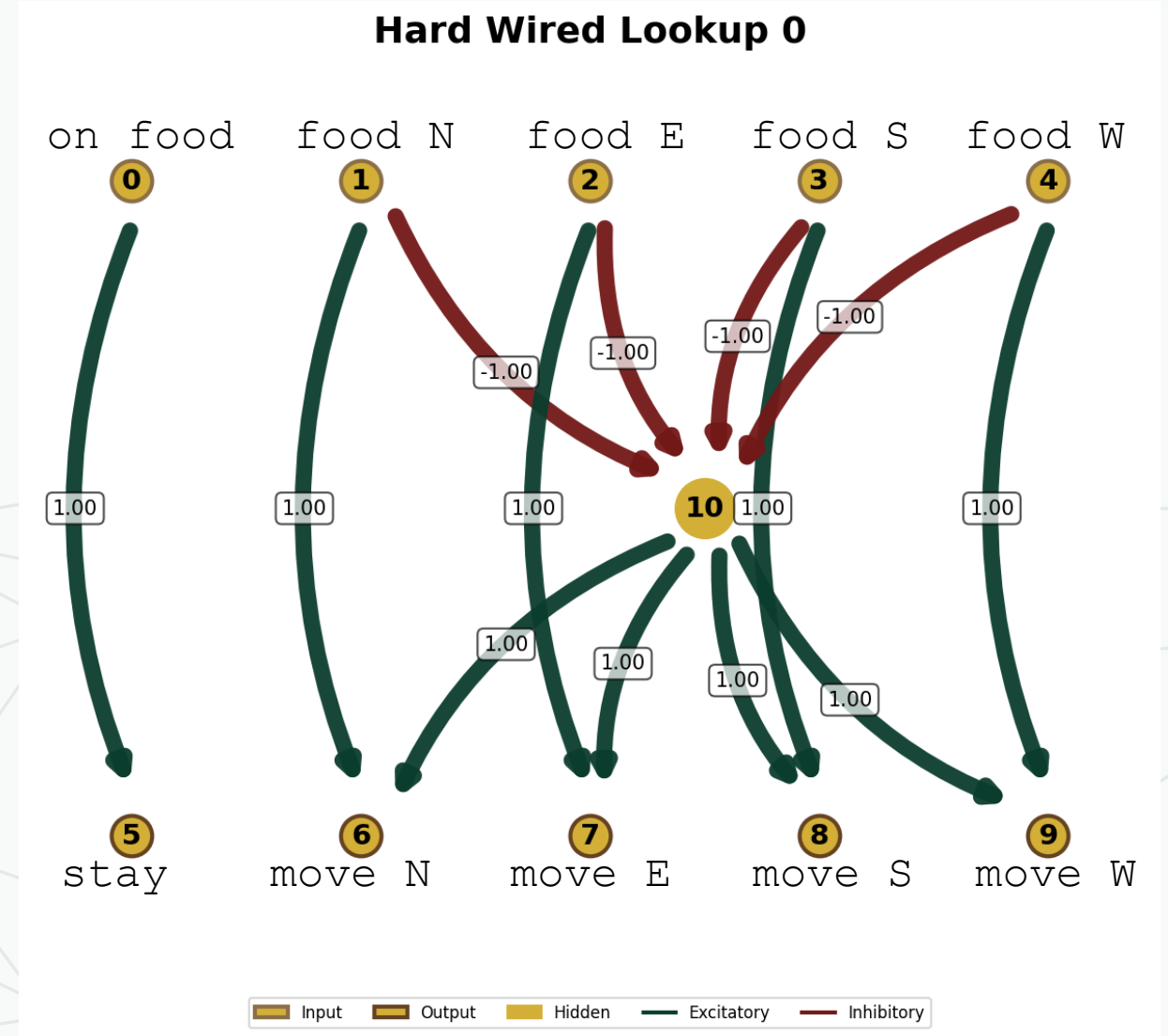
In generation 0: mean = 19.5, median = 19.2, max = 29.7
In generation 50: mean = 69.0, median = 74.22, max = 131.59

Optimal solution?

If on food → stay and eat

If sensing food in one direction → move in that direction

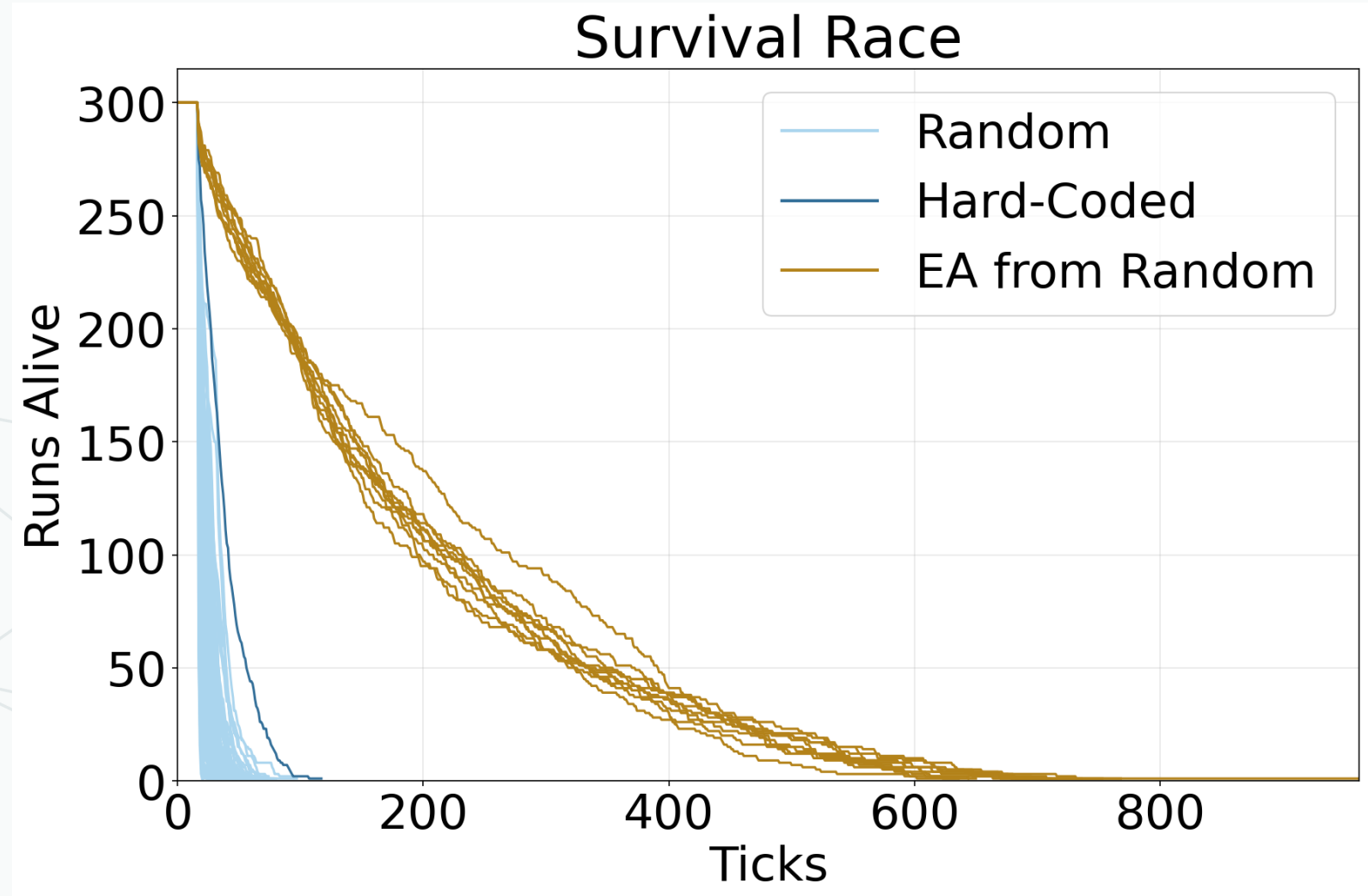
If not sensing any food → move away in random direction



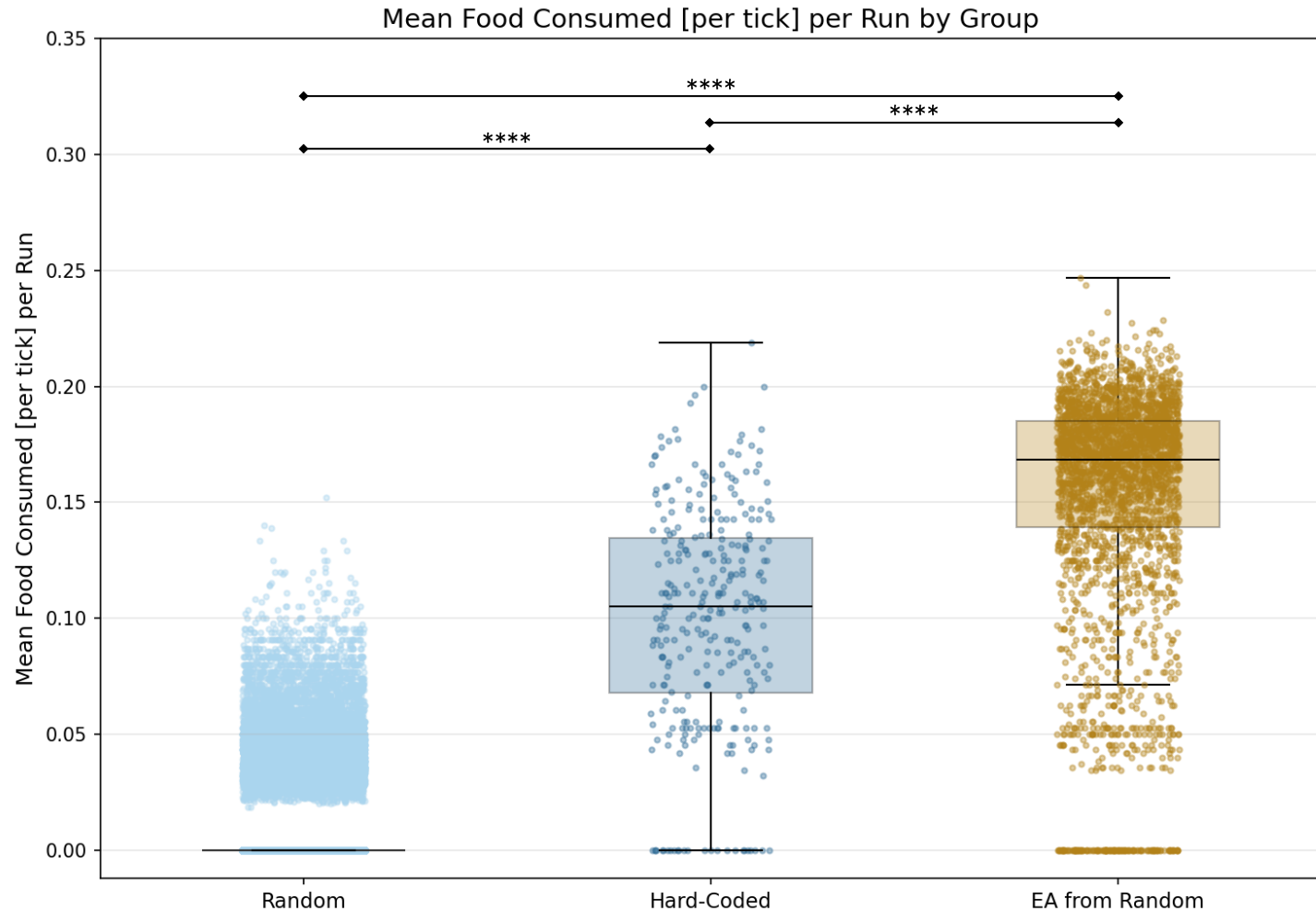
Optimal solution?

No!

Why not and what did the
evolutionary algorithm find?



The evolved variants found
15 times more food than the
random variants!



Kruskal-Wallis: $p < 0.01$; $\epsilon^2 = 0.4$ (large effect size)

MWU + Holm-Bonferroni:

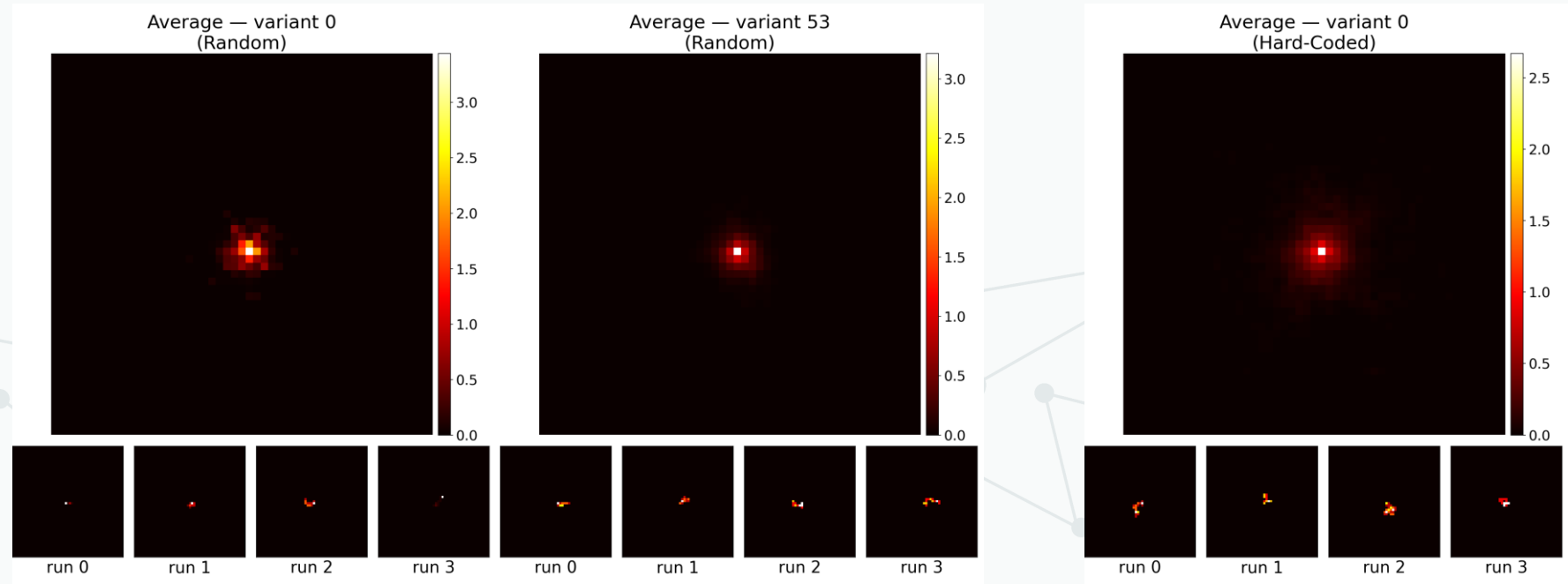
Hard-Coded vs Random: $p < 0.01$; $r = -0.87$ (large effect size)

EA vs Random: $p < 0.01$; $r = -0.97$ (large effect size)

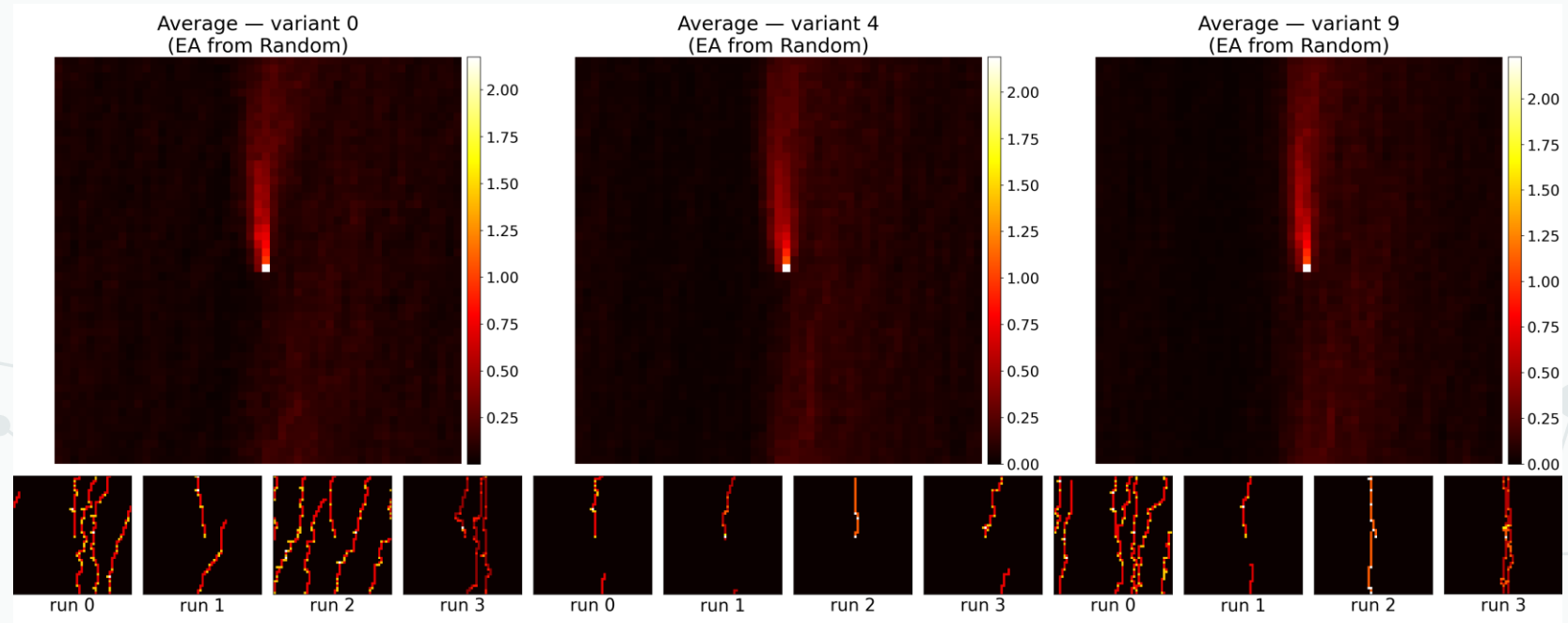
EA vs Hard-Coded: $p = 9 \times 10^{-90}$; $r = -0.7$ (large effect size)

In this scenario food once consumed does not regrow.

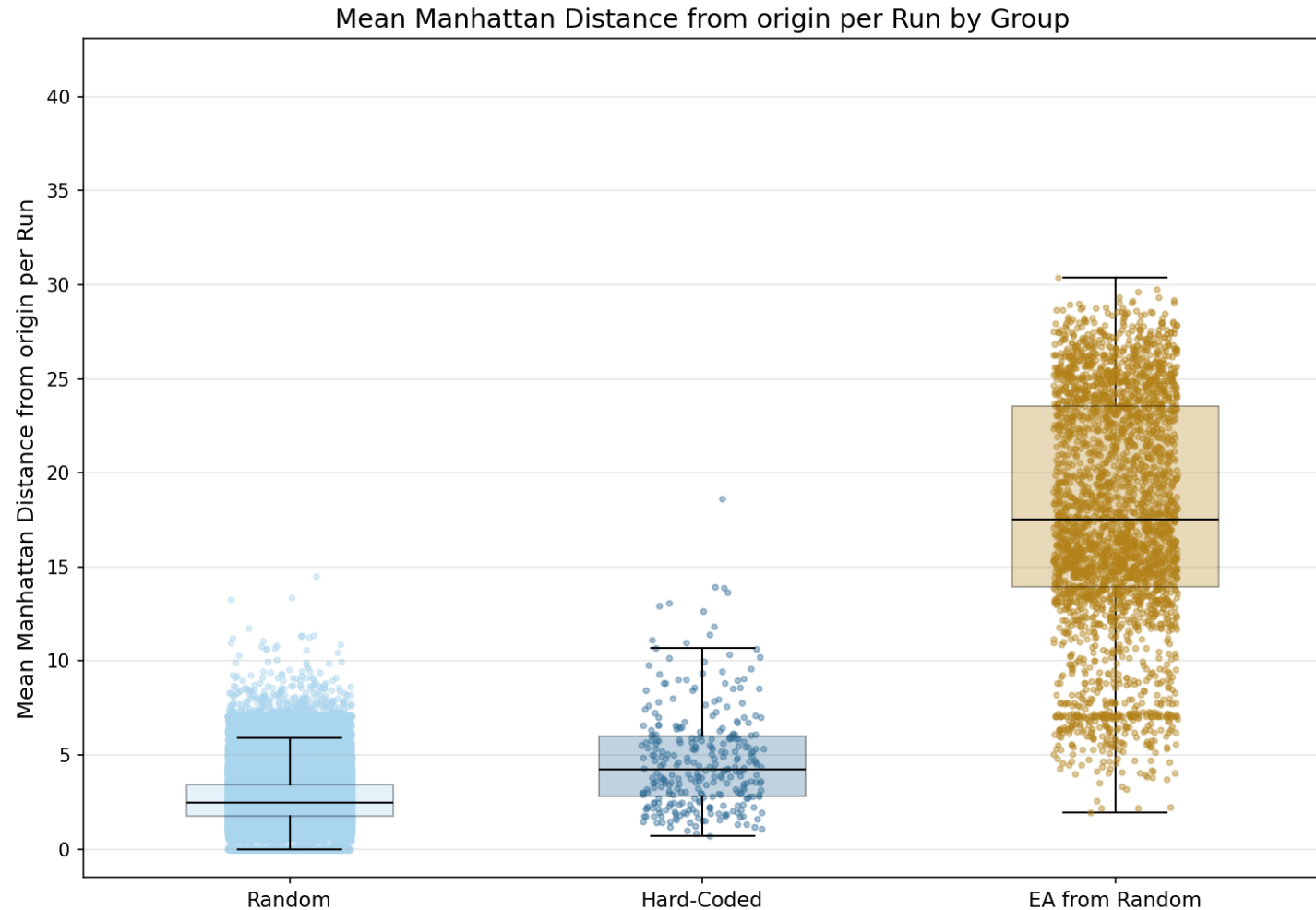
Randomly moving variants deplete their local supply.



Evolved variants developed
a **directional preference**.



The evolved variants **travel further from the origin** than the random variants!



Kruskal-Wallis: $p \approx 0.000$; $\epsilon^2 = 0.1075$ (medium effect size)

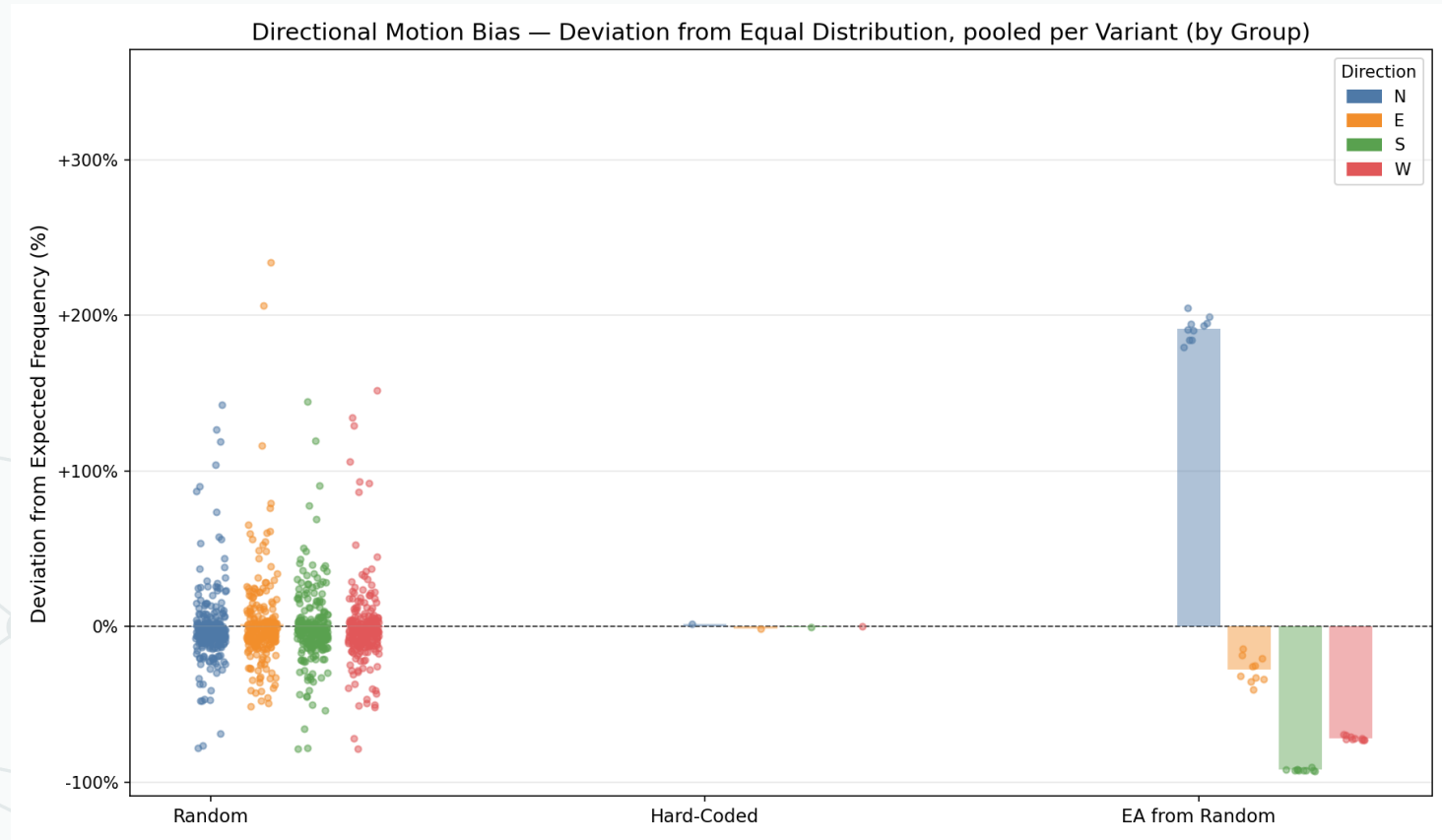
MWU + Holm-Bonferroni:

Hard-Coded vs Random: $p = 1 \times 10^{-50}$; $r = 0.499$ (large effect size)

EA vs Random: $p \approx 0.000$; $r = 0.992$ (large effect size)

EA vs Hard-Coded: $p = 1 \times 10^{-159}$; $r = 0.943$ (large effect size)

Evolved variants developed
a **directional preference**.



Random:

Kruskal-Wallis $p = 10^{-40}$; $\epsilon^2 < 0.01$ (n.s. ES)

MWU + Holm-Bonferroni:

N vs. E: $p \sim 0.1$; $r < -0.01$ (n.s. ES)

N vs. S: $p \sim 0.6$; $r < -0.01$ (n.s. ES)

N vs. W: $p \sim 4 \cdot 10^{-29}$; $r \sim -0.05$ (n.s. ES)

E vs. S: $p \sim 0.003$; $r \sim -0.01$ (n.s. ES)

E vs. W: $p \sim 2 \cdot 10^{-20}$; $r \sim -0.04$ (n.s. ES)

S vs. W: $p \sim 3 \cdot 10^{-32}$; $r \sim -0.06$ (n.s. ES)

Hard-Coded:

Kruskal-Wallis $p = 0.65$; $\epsilon^2 < 0.01$ (n.s. ES)

EA from Random:

Kruskal-Wallis $p < 0.001$; $\epsilon^2 = 0.6$ (large ES)

MWU + Holm-Bonferroni:

N vs. E: $p < 0.001$; $r = -0.7$ (large ES)

N vs. S: $p < 0.001$; $r = -0.99$ (large ES)

N vs. W: $p < 0.001$; $r = -0.92$ (large ES)

E vs. S: $p < 0.001$; $r = -0.7$ (large ES)

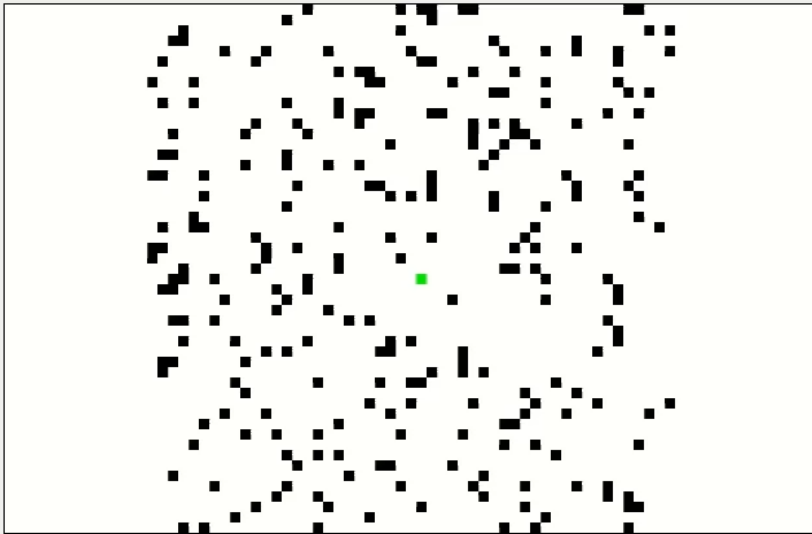
E vs. W: $p \sim 6 \cdot 10^{-142}$; $r = -0.4$ (medium ES)

S vs. W: $p < 0.001$; $r = 0.6$ (large ES)

Evolutionary algorithm finds successful solutions to **static** tasks.

Can these adapt to run-time rule changes?

Can these adapt to run-time rule changes?



SENSE | on_food:0 food_north:0 food_south:0 food_west:0 food_east:0

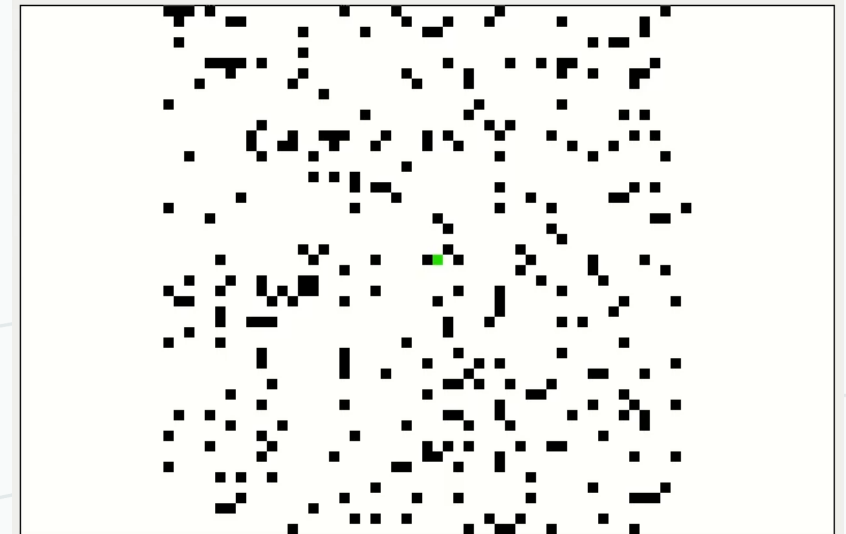
tick=0 energy=30 eats=0 dist=0 ALIVE

Initially:

dense targets,
no regrow

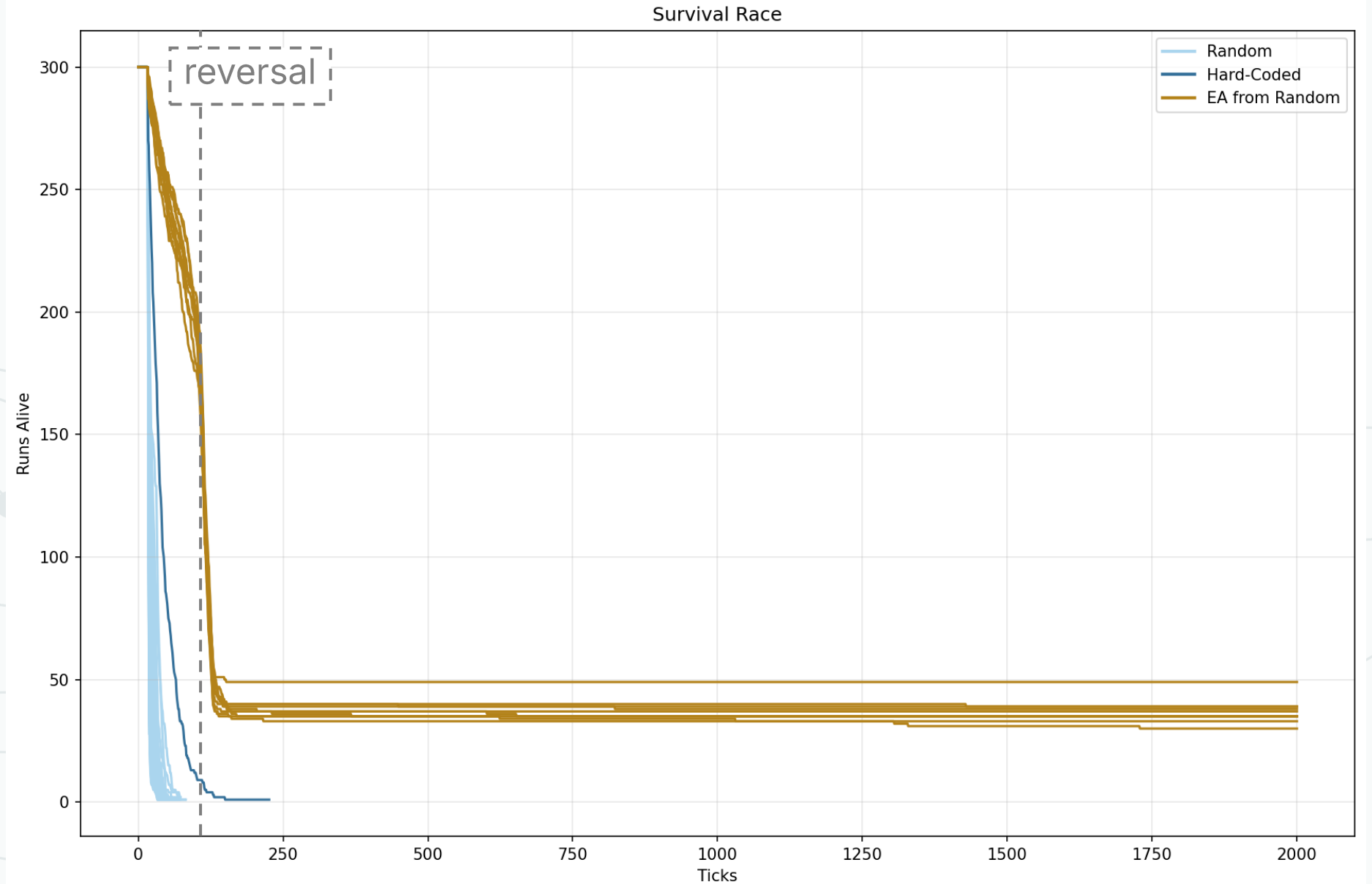
Tick 100:

sparse targets,
fast regrow



SENSE | on_food:0 food_north:0 food_south:0 food_west:1 food_east:0

tick=2 energy=26 eats=0 dist=2 ALIVE



Evolutionary algorithm finds successful solutions to **static** tasks.

Can these adapt to run-time rule changes?

Yes!

1

Static Task Phase

Train on static task
w/ traditional
evolution and define
adaptability gap

2

**Training & Testing EA
with perturbations**

Train variants w/
perturbed evolution and
test performance and
adaptability

3

Ablation

Ablate to necessary
core and test
adaptability on withheld
scenarios

4

Wiring analysis

Analyze common
patterns in adaptors and
compare with biological
wiring patterns

Outlook

Build minimal networks that can adapt to tasks they were never explicitly trained on, and characterize what structural properties make that possible.

- Candidate wiring motifs for targeted investigation in biological circuits / networks
- Model for interpretation of data from complex circuits and brains